

AI & Climate+ Sustainability Sandbox

Enhancing Land-Use Segmentation for ODOS Tech

Chantelle - Dáire - Feng - Lainey - Prabha - Purity



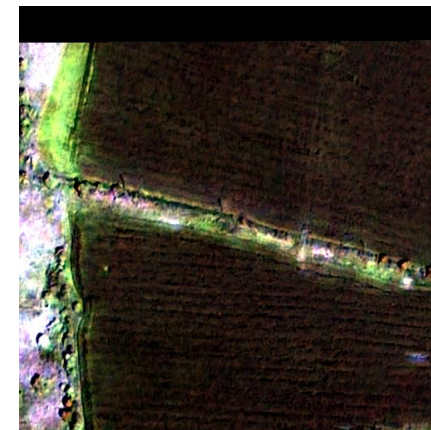
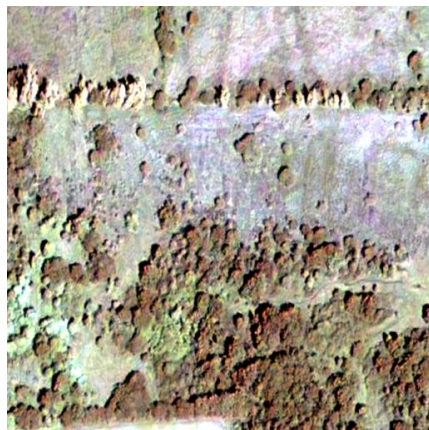
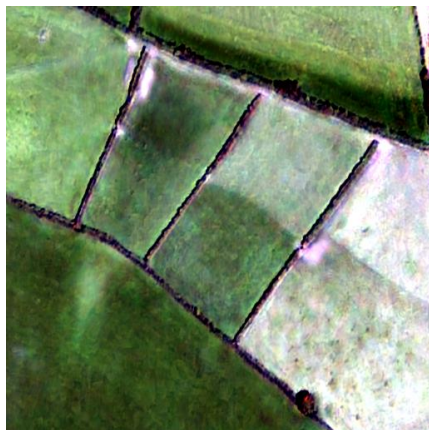
Overview



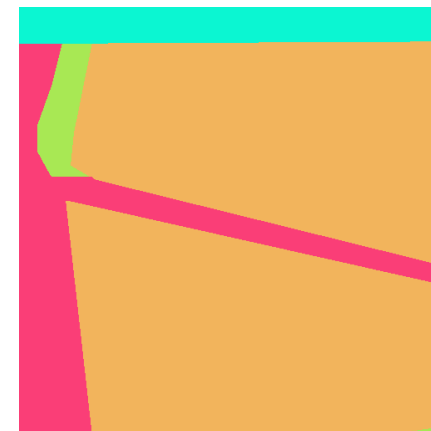
- I. Challenges
- II. Objectives
- III. Data
- IV. Exploratory analysis
- V. Methods:
 - Data Augmentation & Resampling
 - Model Architecture
- VI. Results
- VII. Additional Experiments
 - Semi-Automated Mask Generation
 - Generative Data Augmentation
- VIII. Recommendations

Data

Satellite
images



Masks



 Forest land  Cropland  Seminatural Grassland  Grassland  Background  Settlement

Dataset: There are **2,307** image/mask pairs, which were **made by hand**!

Challenges



1 **Technical** → Outdated method → 85% mIoU plateau

2 **Operational** → Manual editing → slow & inconsistent



Satellite Image →



Outdated Image Analysis →



Manual Editing →



Labelled Map



What has been done so far

ODOS have tested several computer vision model architectures:

a) UNetFormer,

b) FT-UNetFormer (used as our baseline model),

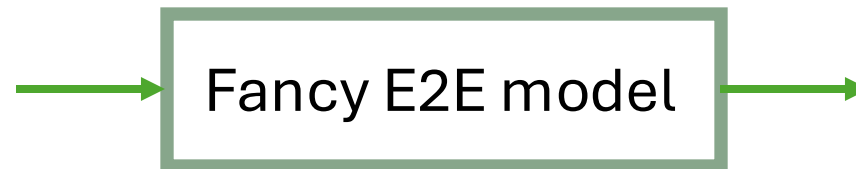
c) and DCSwin.

Main Objective

We want to predict the mask for any new satellite image.



Input:
Unseen satellite image



Output:
Segmentation mask

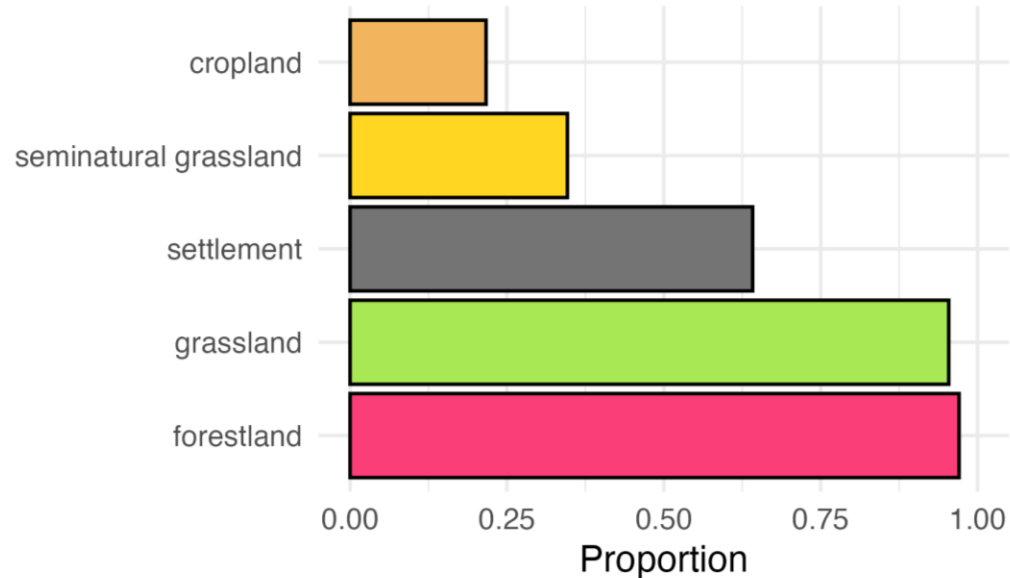
Mapping Goals to Impacts

 Goals →  Impact

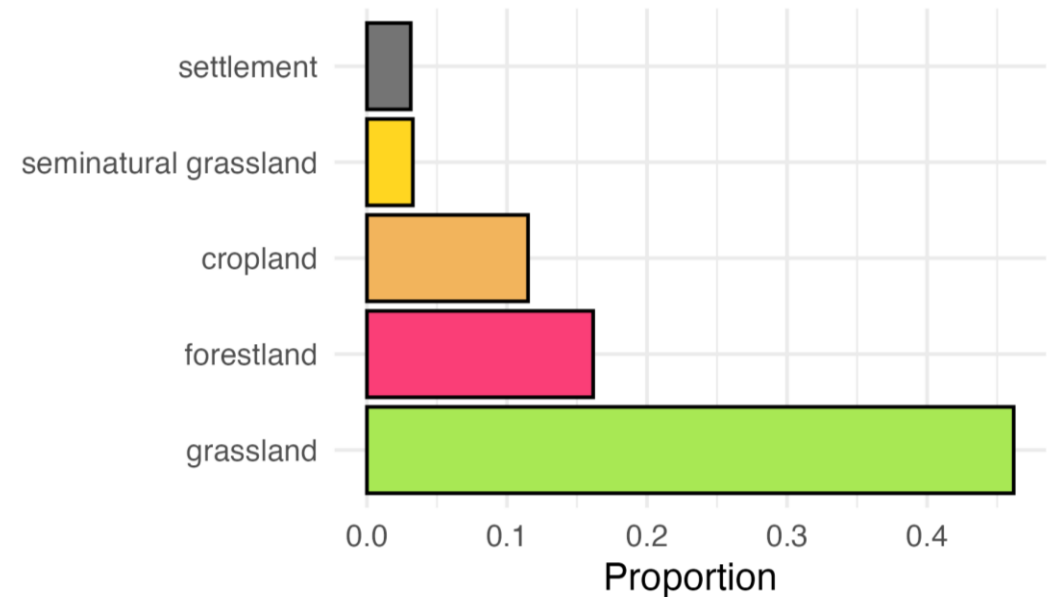
- | | | |
|---|---|------------------------------|
| 1 Diagnose & balance data | → | Improved class balance |
| 2 Enhance model architecture | → | Better performance (mIoU ↑) |
| 3 Automate / generate masks & data | → | Faster, more consistent data |
-

Class imbalance

Proportion of images containing each class



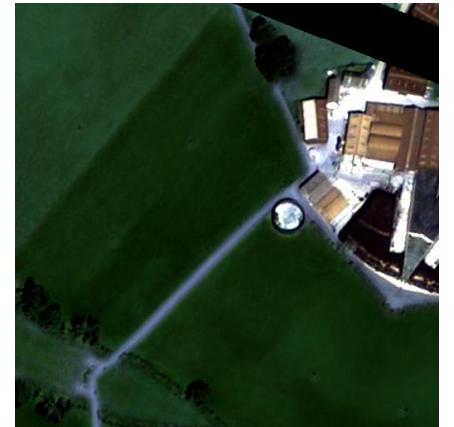
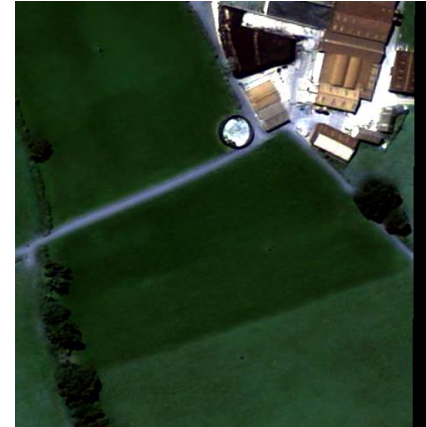
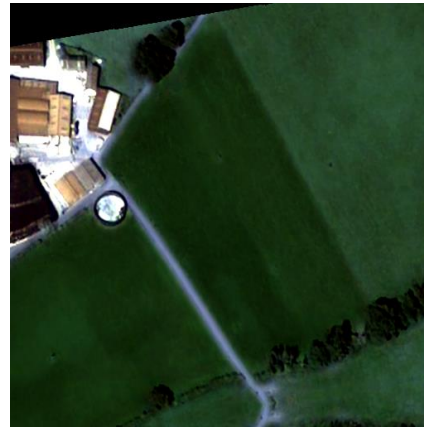
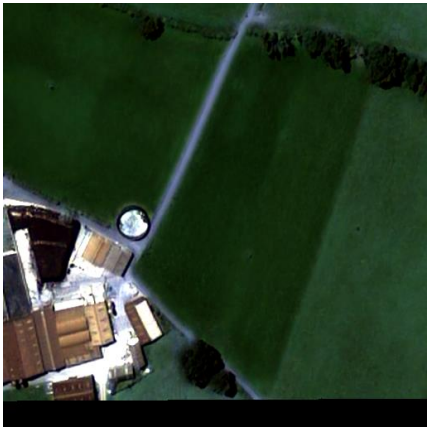
Mean proportion of pixels containing each class



- Most images contain forest, grassland and settlement
- Grassland dominates most images in terms of pixel count.
- Some classes take up very few pixels even when present.
- This is most problematic for rare classes.

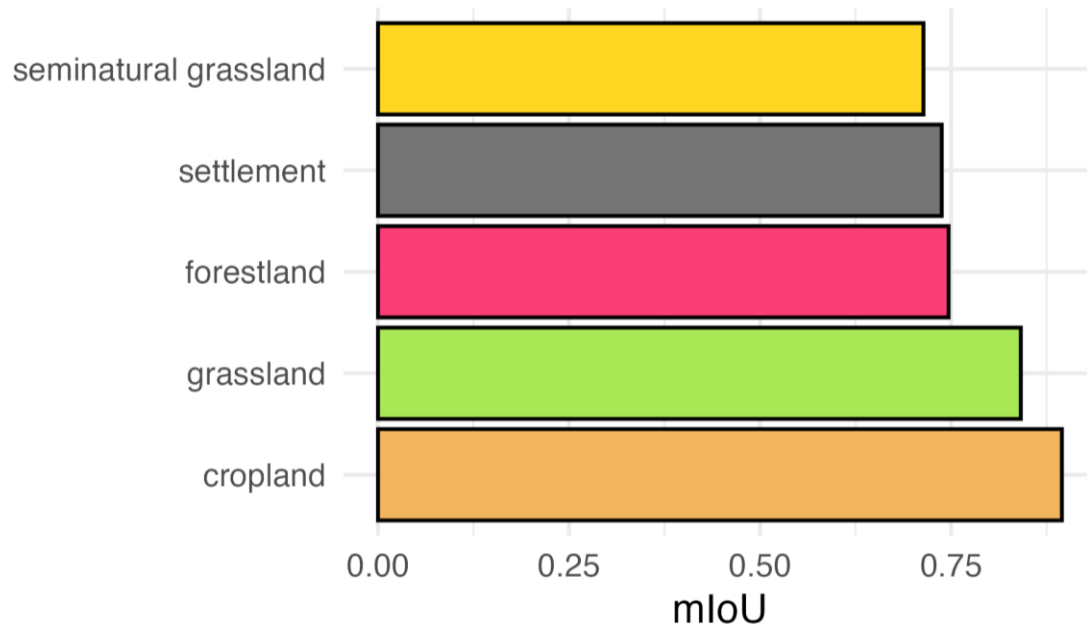
Data augmentation

- We devise an algorithm to identify which classes were underrepresented in the dataset .
- We resample images containing these rare classes based on an importance score.
- From 2307 images, 1615 images were resampled and augmented, 2271 training images(+~40%)



Model performance by class

Base model (FT-UNetFormer) performance by class



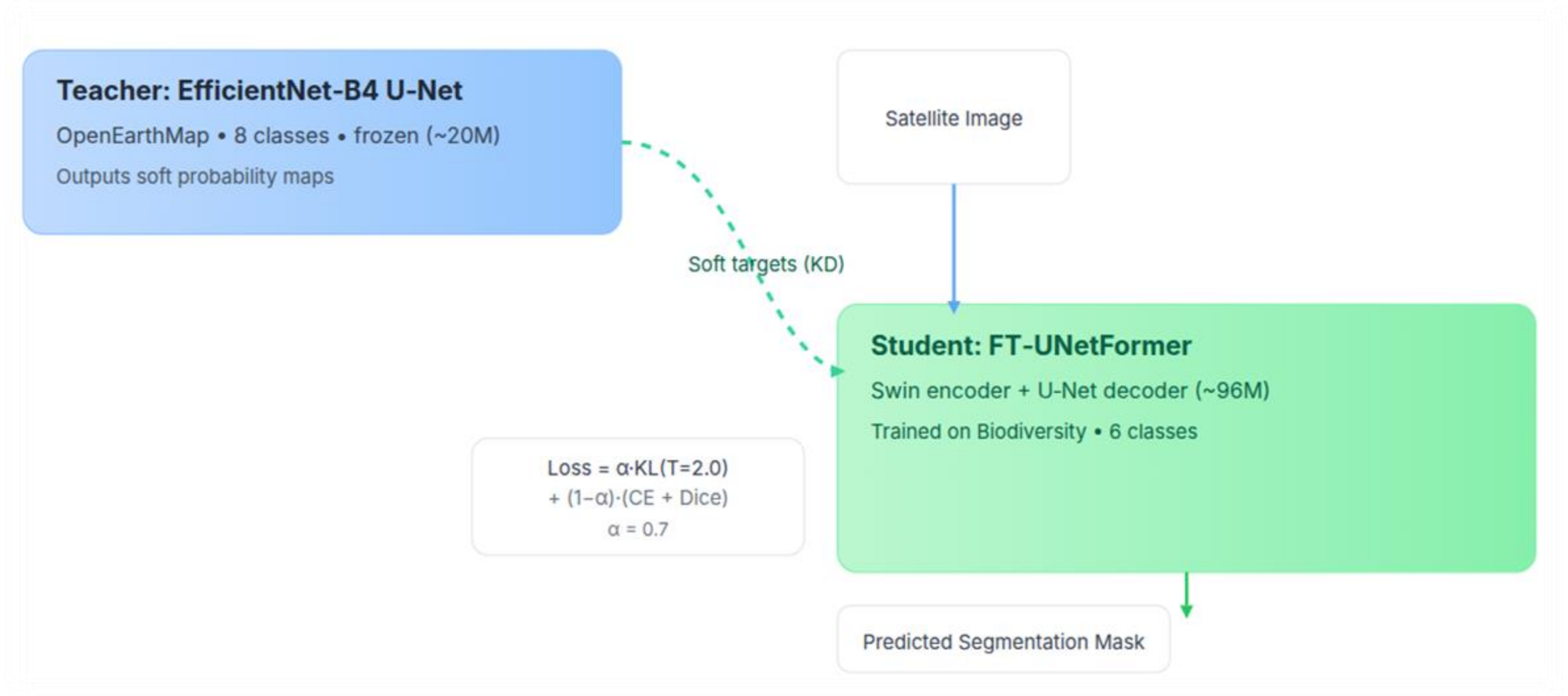
Model does worst at segmenting the rarer classes

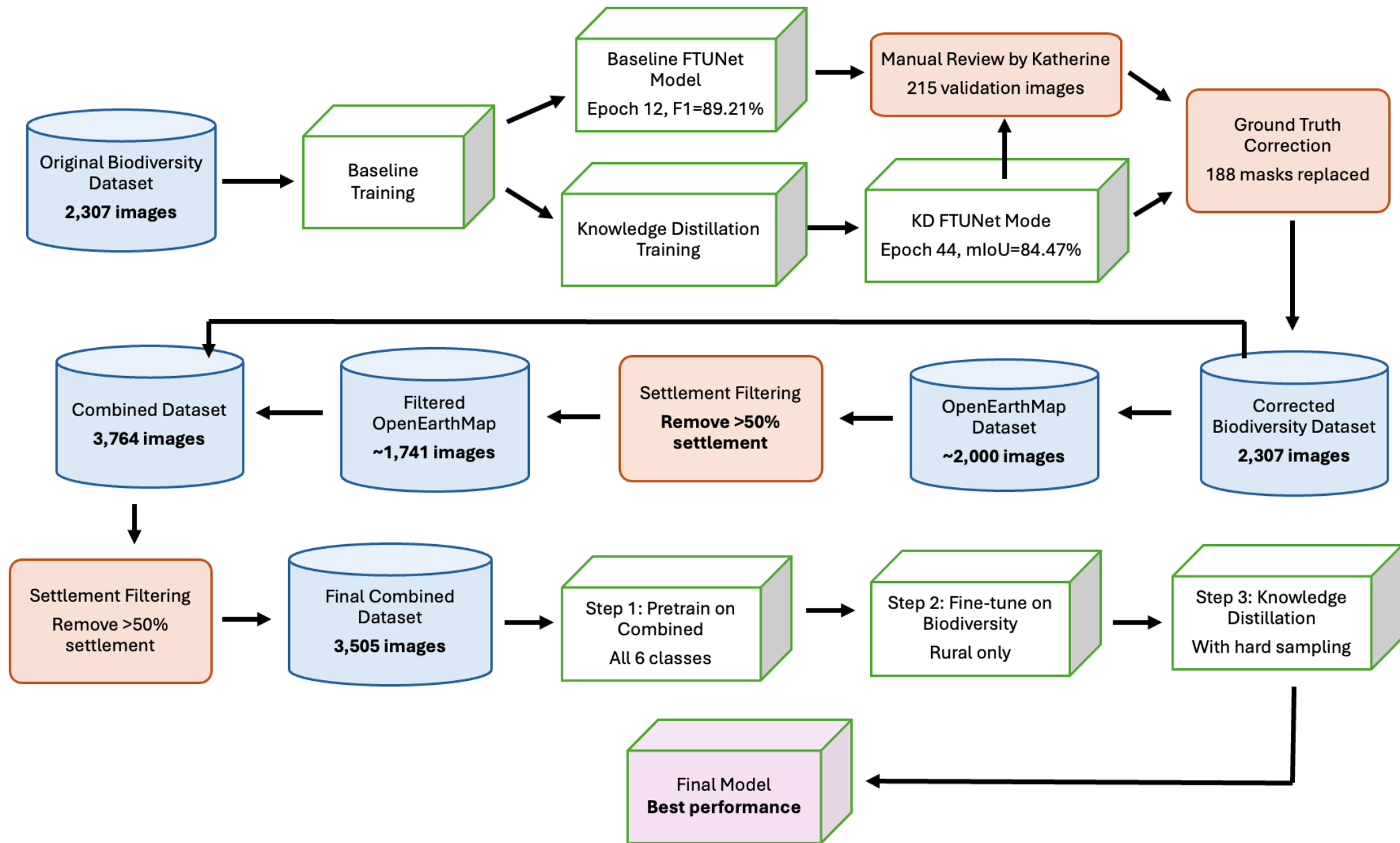
We found model improvement as follows:

Class	% change in mIoU
seminatural grassland	5.36
settlement	4.85
cropland	-1.37
grassland	0.118
forestland	2.10

Model Architecture – FT – UNetFormer + Knowledge Distillation

Teacher-Student setup leveraging EfficientNet-B4 U-Net trained on OpenEarthMap



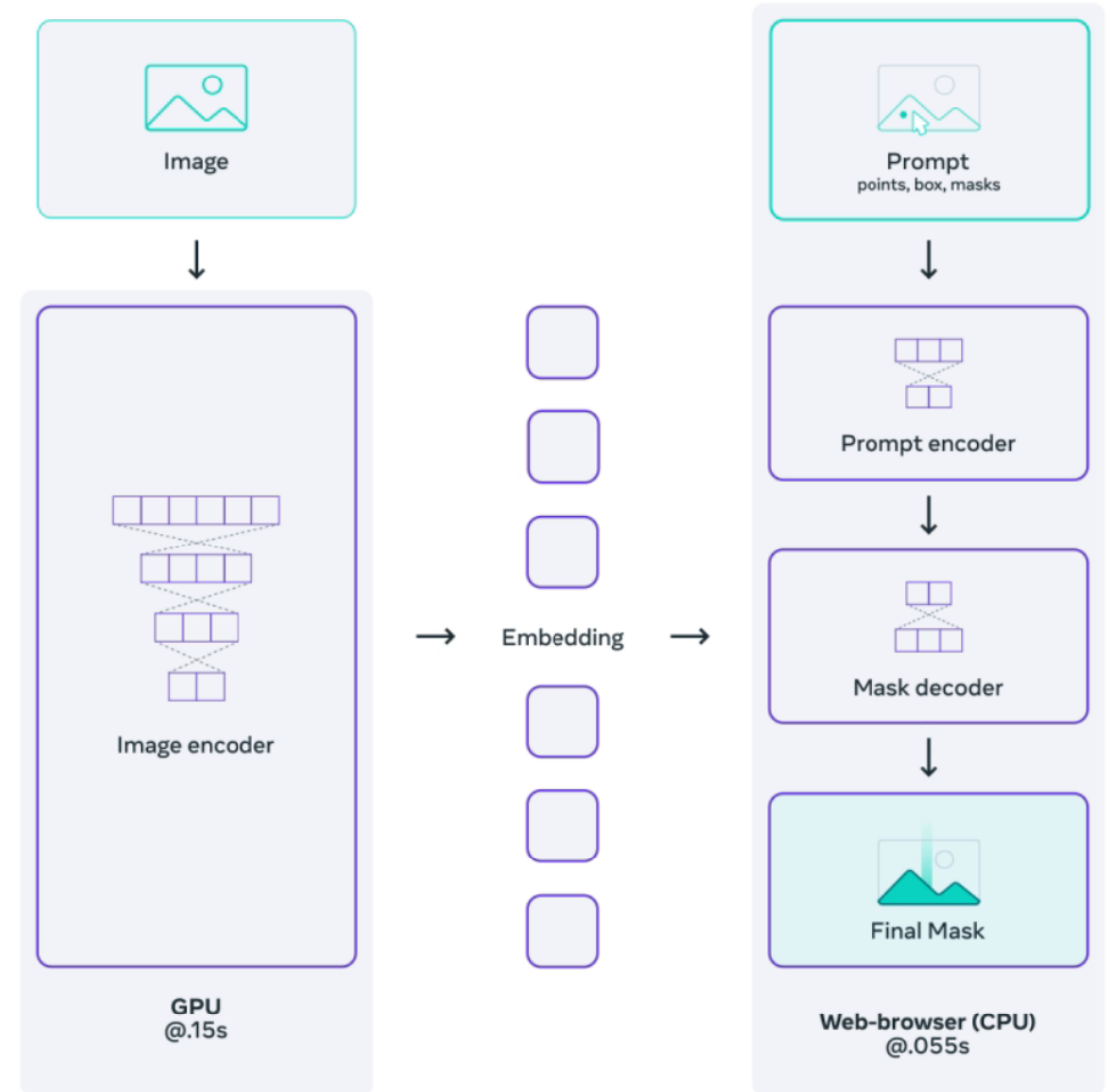


Results

Training Run	Dataset Used	mIoU
Phase 0: Baseline FTUNetFormer	2,307 original images	~72%
Phase 1: + Knowledge Distillation	2,307 original images	84.47%
Phase 1b: KD + TTA (test)	Same as Phase 1	83.93%
Phase 6: Full Augmentation	2,271 replicated images	87.57%
Phase 7: Phase 6 + TTA (rejected)	Same as Phase 6	86.63%
Phase 8: + Strategic Cropping	2,271 replicated images	88-89%*

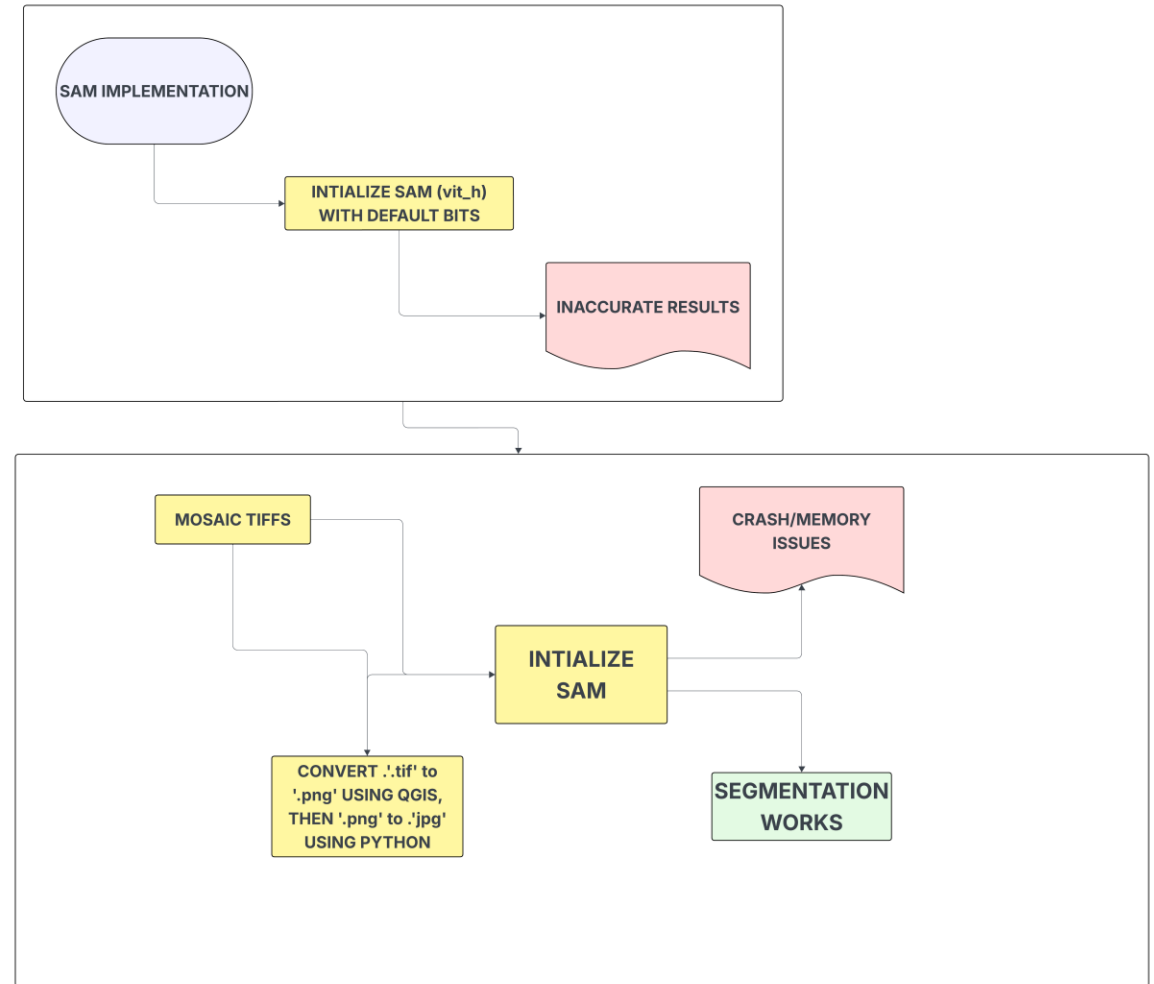
SAM

- Eliminates Manual Mask Drawing
- Adapts Well to Satellite Imagery
- Supports Fully Automatic and Interactive Modes
- High-Quality Pixel-Accurate Segmentation



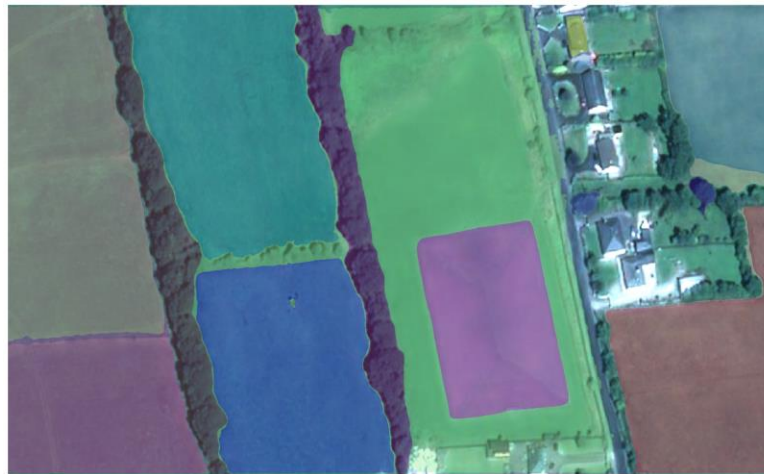
Semi-Automated Mask Generation

- Tried implementing SAM on satellite data.
- Mosaic
- Conversion of .tiff to jpeg



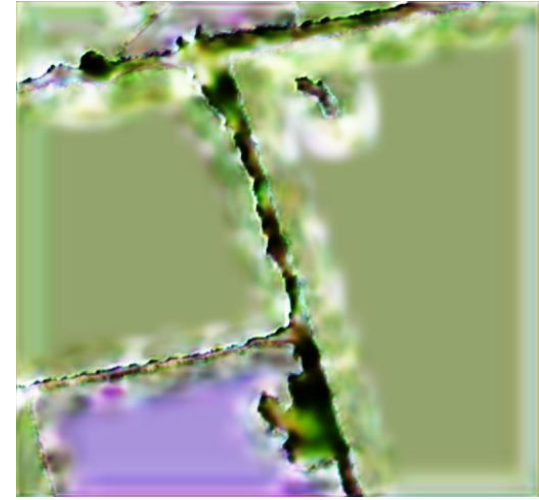
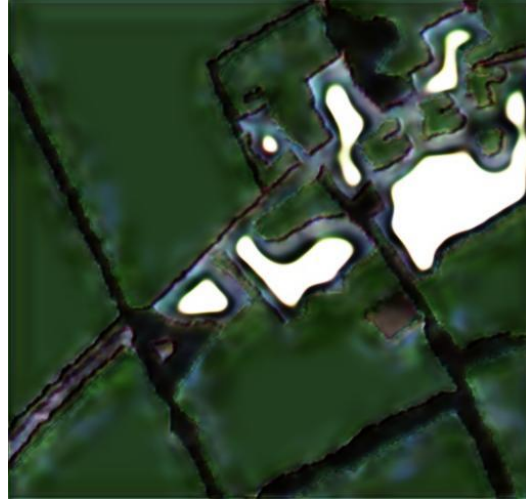
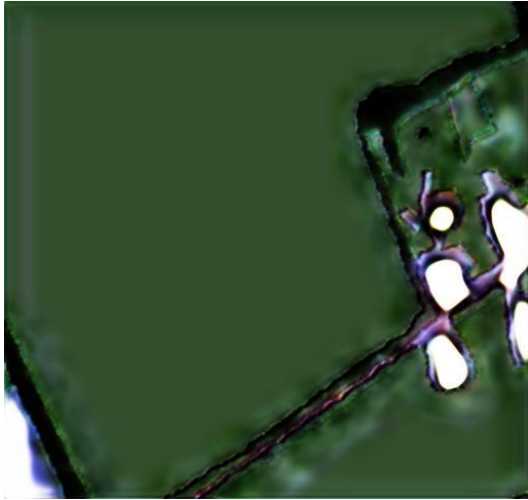
Semi-Automated Mask Generation

Results from
SAM(vit_h)
model



Generative Data Augmentation

GAN-synthetic filtered (100 epochs)



Study Limitations

- Only **3 regions** represented and limited dataset size
- "Ground-truth" masks were manually drawn / partially auto-generated – **mislabeled classes**
- Training limitation due to the devices(i.e., **GPU memory**)

Recommendations

1. Develop the automatic mask generator into a semantic or semi-automated labelling tool
2. Adopt and retrain more fine-tuned foundation-models approach to move toward the 95 % mIoU target.
3. Expand the dataset to new geographic regions with similar climatic zone and comparable land-use types

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